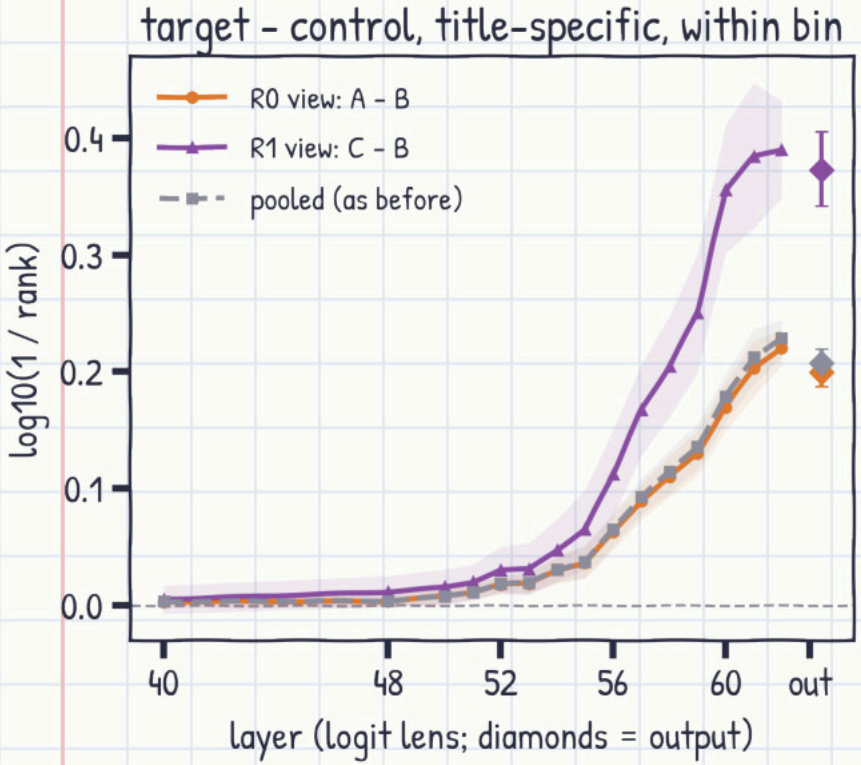
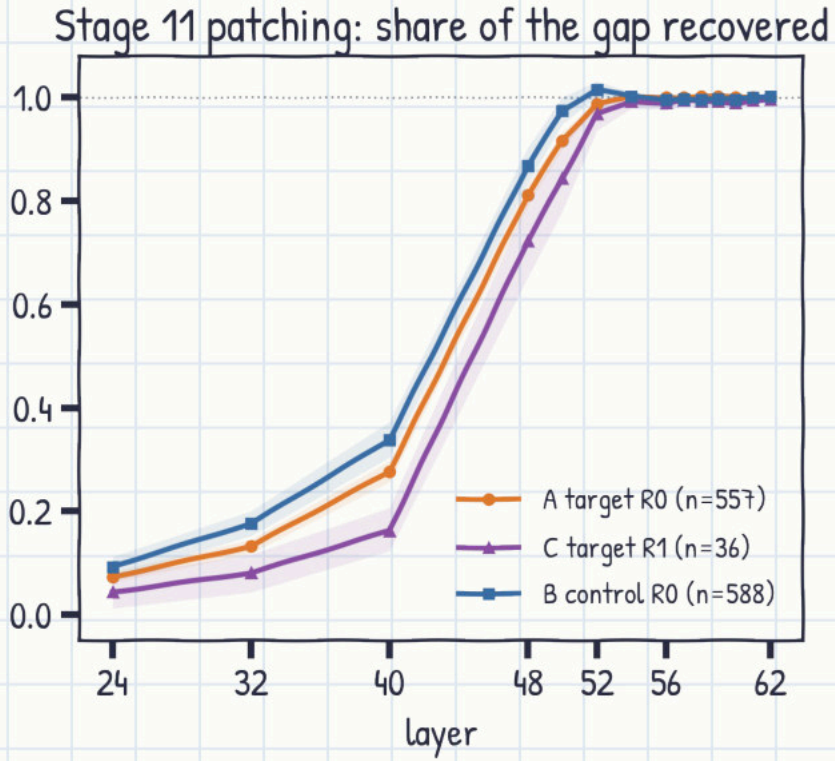
Stage 10–11

Is it just the R1 papers? No.

R0 = no true author written · R1 = a few written · A/B/C/D = target/control x R0/R1





R0 only (A – B):

output +0.199 vs +0.207, same shape (Stage 10, A 3,423 / B 2,151, §6.13)

R1 (C – B): +0.37, also late (C 391 / B 885)

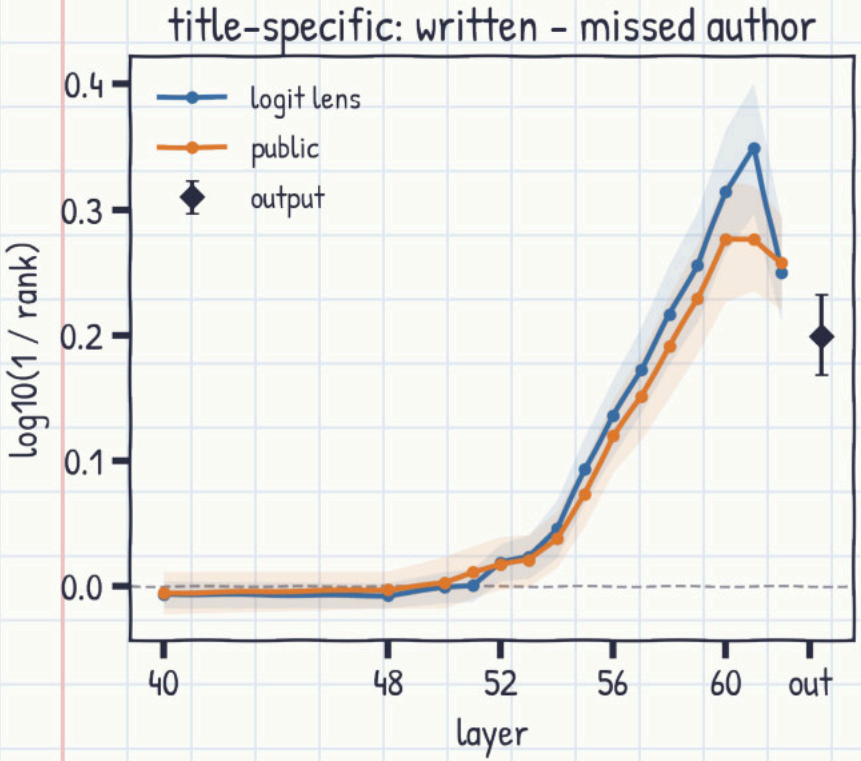
C – A within target: +0.11, from L56 (C 485 / A 3,204)

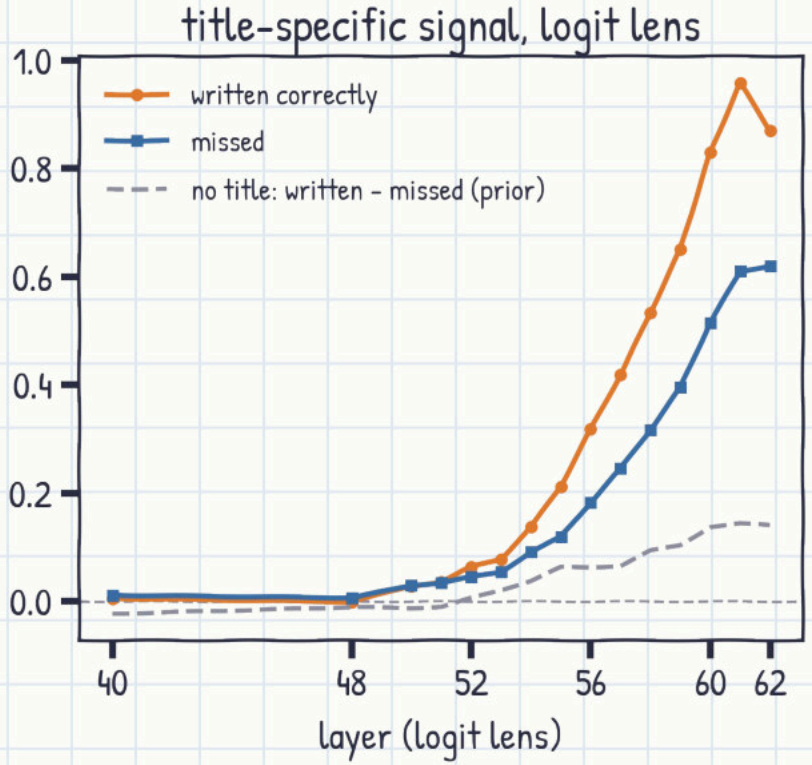
patching: R1 a bit later, done by L52 (Stage 11 E1, 36 R1 targets)

Stage 10–11

Authors it wrote vs authors it missed

inside 495 R1 papers · paired within paper · 95% CI (Stage 10, r0_r1_analysis.md §4)





written − missed, output:
+0.20 (Stage 10, final L63, 495 papers)

~0 up to L52, grows from **L54**

no title (prior): +0.10 (same, no title)

same without first authors

Stages 1–9 · 20–200 papers

What we saw

- True title helps: pilot L56 +9.3 pp(7.3, 50 papers); 200 new papers +5.2 pp(8D RQ1, 200 papers)
- A–B uncertain in the pilot; +5.0 pp on 200 papers but gone after dropping 10 pairs(8D RQ2, 100 pairs)
- Low – high citation: inconclusive

Stages 10–11 · 9,108 papers

What scale showed

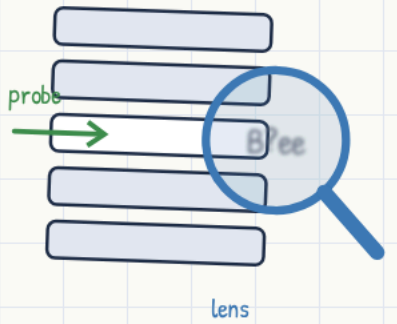
- Title-specific signal clear in every group, ≈0.44 at L62(L62, §6.9)
- target – control ≈0 at L24–L40, slightly negative at L48 (−0.04 to −0.05)(§6.4), +0.21 [0.20, 0.22] at the output(L63, §6.4)
- Name prior (delete-title condition) is only ~0.35 of the gap; ~0 on surnames(§6.12)
- Signal rises steadily with citations (ρ = 0.451)(§6.8)
- Same groups as the pilot (R0 only, A – B): +0.20 [0.19, 0.21] (R0, §6.13), same late shape
- R1 papers (C) know more (+0.11 over A)(§6.13) but nearly as late

Line A: from "the title helps" to "this paper, late, graded by fame"

The pilot was right that the title matters. It could not see that the target advantage is real but confined to the last layers, so the 7-layer average and the L56 endpoint both diluted it. Restricting the full cohort to the pilot's A/B groups (R0 only) changes nothing: the R1 papers do not drive the result.

Line B

What can each readout see?



small samples asked

Is J-Lens better than Vanilla? Is the fitted lens stable?

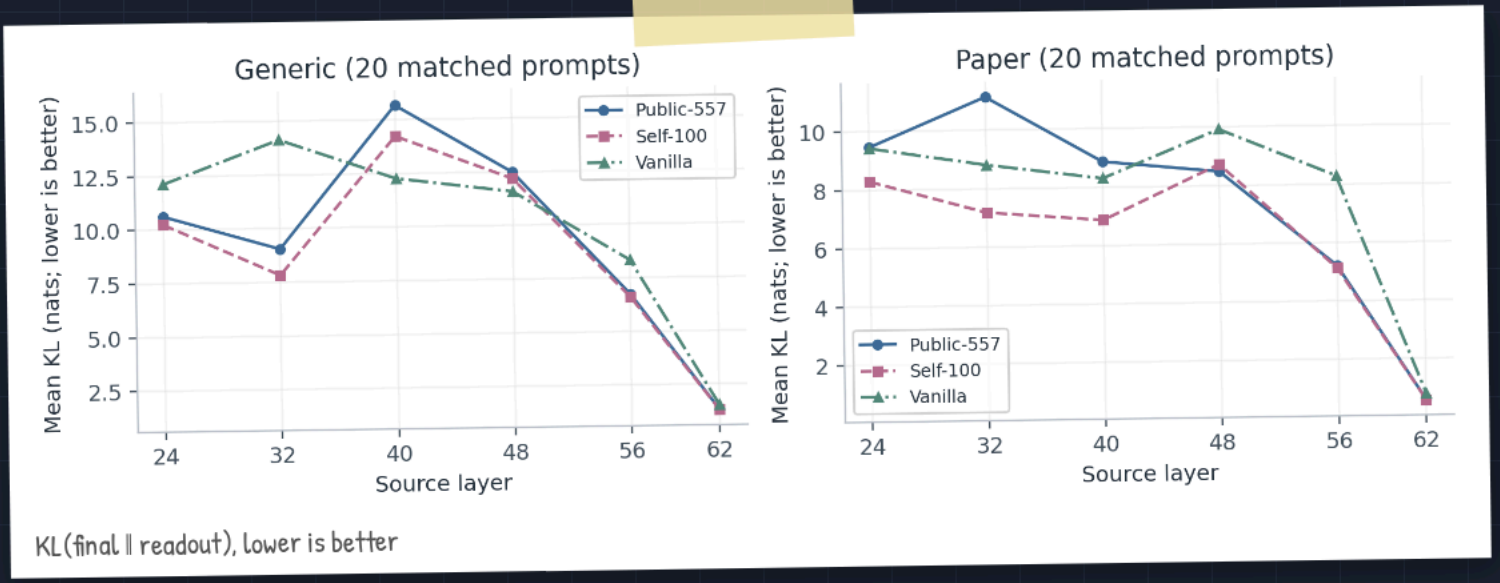
→

the full cohort answered

Lenses read name shape earlier, not paper identity; a probe at the title token reads paper-specific information

Two lenses: Public-557 vs Self-100

Can a middle layer approximate the final next-token distribution? (20 generic + 20 paper prompts)

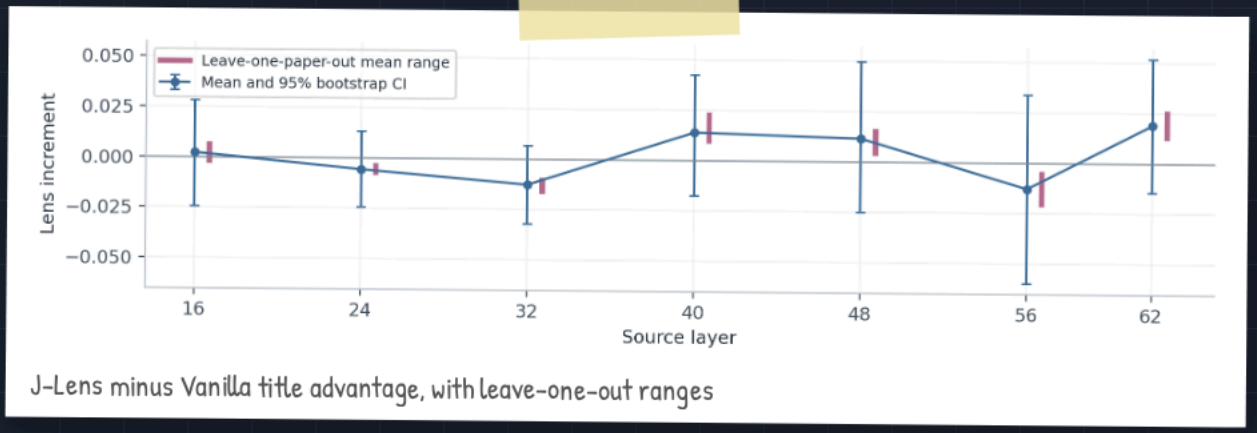


- Both checkpoints numerically valid (63/63 and 6/6 finite)(matrices.csv; frobenius_norms.csv)
- Self-100 has lower KL on 6/6 generic and 5/6 paper layers (paper L32: 11.1 → 7.2) (20 + 20 prompts; metrics.csv)
- But at paper L62, Self-100 KL beats Vanilla (0.58 vs 0.81) while Top-10 overlap is worse (0.36 vs 0.72)(metrics.csv)
- Author pilot: both lenses 14/20 full-name wins at L48(20 papers, layer_summary.csv)

Lesson: a better distribution match does not mean better author ranking. Fidelity cannot stand in for the task.

Cross-stage audit: does J-Lens add value?

Read-only joins of Stages 5–7.3; 20,000 bootstrap draws



J-Lens minus Vanilla title advantage, with leave-one-out ranges

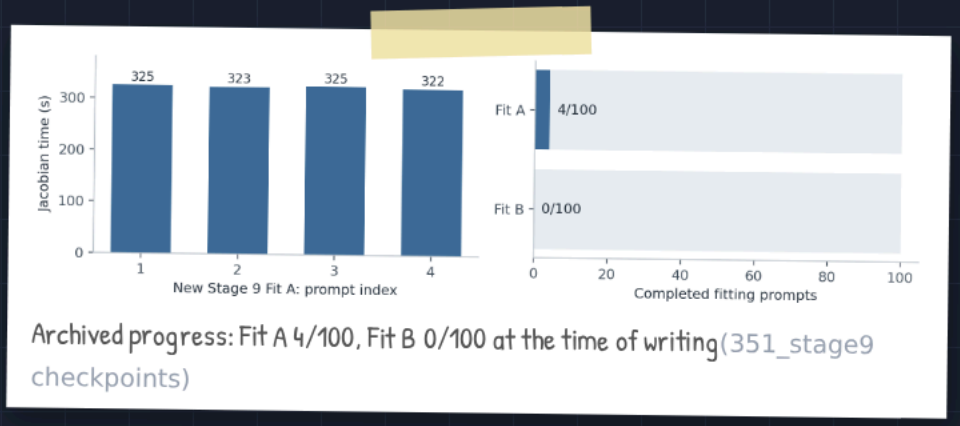
Stage 5 vs 7.3 at L56	r = 0.45, ρ = 0.47; signs agree 32/50 (50 papers, cross_stage_correlations.csv)
J-Lens − Vanilla	+0.002 [−0.008, +0.012]; median −0.002; 24/50 positive (jlens_increment_bootstrap.csv)
A−B of that increment	−0.014 [−0.033, +0.004] (20 pairs)

Dropping one paper can flip the sign of the J-Lens increment. The pilot shows **no added title-specific value** from J-Lens over Vanilla. Everything so far is exploratory: L56 was inspected repeatedly.

Stage 9: is the lens stable across fitting data?

New FullFit protocol, different from the historical 50+50 split

	Historical split-half	Stage 9 FullFit
Fit data	Self-100 split 50 + 50	two disjoint 100-prompt sets
Holdout	20 generic + 20 paper	independent 100 WikiText
Layers	6	7 (adds L16)
Status	complete	only Fit A completed; Fit B and holdout not run



Archived progress: Fit A 4/100, Fit B 0/100 at the time of writing (351_stage9 checkpoints)

- WikiText-103 train, 100–1,200 chars, deduplicated by hash, seed 3519001
- ~324 s Jacobian time per prompt, ~62 GiB CPU RAM (full_A.log; benchmark log); per-prompt checkpoints

Only Fit A was finished, and Stage 10 used it as "fitA". Without Fit B and the holdout, fitting stability is **untested**.

Layer-by-layer readout

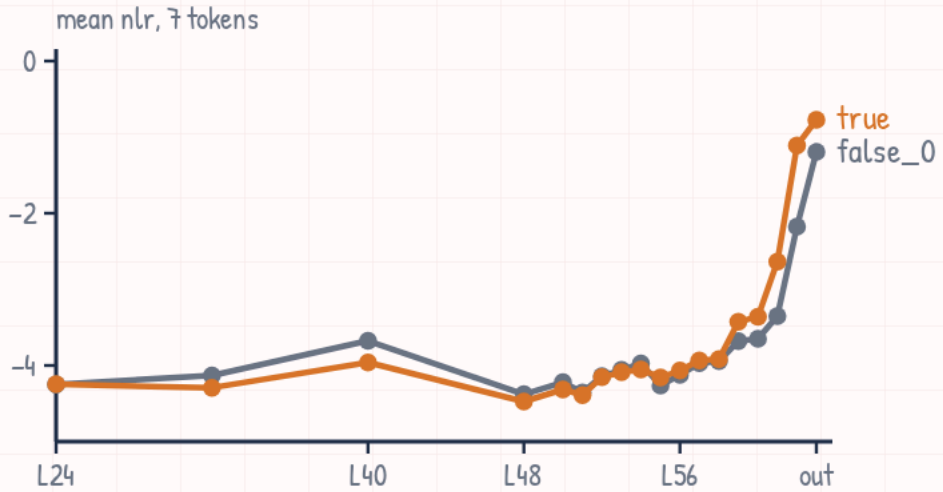
Coifman, Bonanno, Rafaeli (2007) · logit lens

1 prompt → 2 state at name → 3 lens → vocabulary → 4 rank of true token → 5 mean

nlr = log10(1 / rank)

rank = 1 + #{tokens scored above the true one}

怎么算：nlr = log10(1/rank) = −log10(rank)；rank = 1 + #(logit > gold logit) · 1 篇示例，“Co”，logit lens · 出处：raw/logit/L*/shard_011.npz 第 25029 行



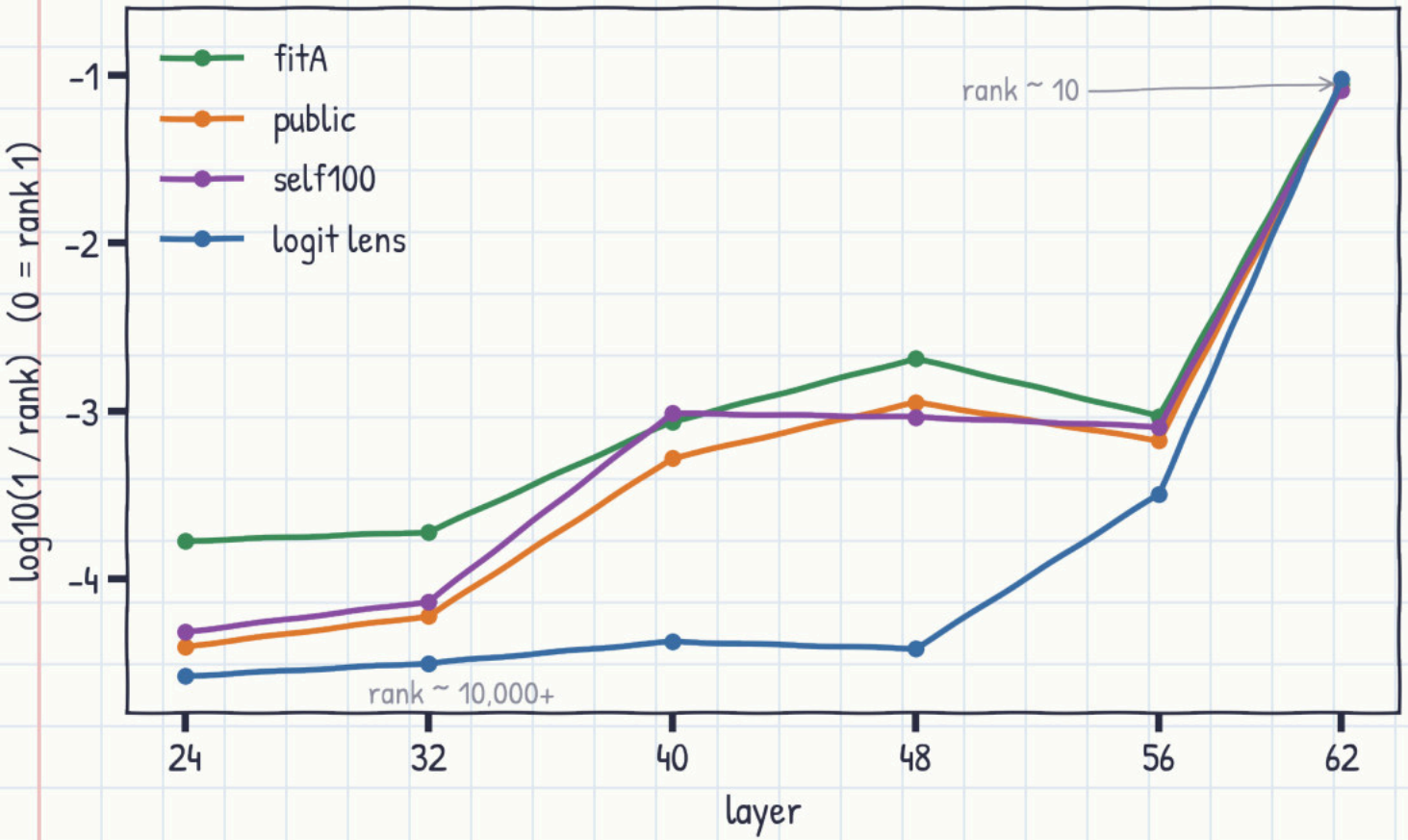


…titled | Affect | dynamics, | … | (2007)? | Authors | : | Kar | in | G | . | Co | if | man

Ranks of “Co” (author 1, token 5), true title: 351_stage10/raw/logit/L{24…62}/shard_011.npz row 25029; out = raw/final/L63

Names become readable late

all 9,108 papers · true title (Stage 10 cohort, §6.1)



Logit lens: almost nothing until **L48**

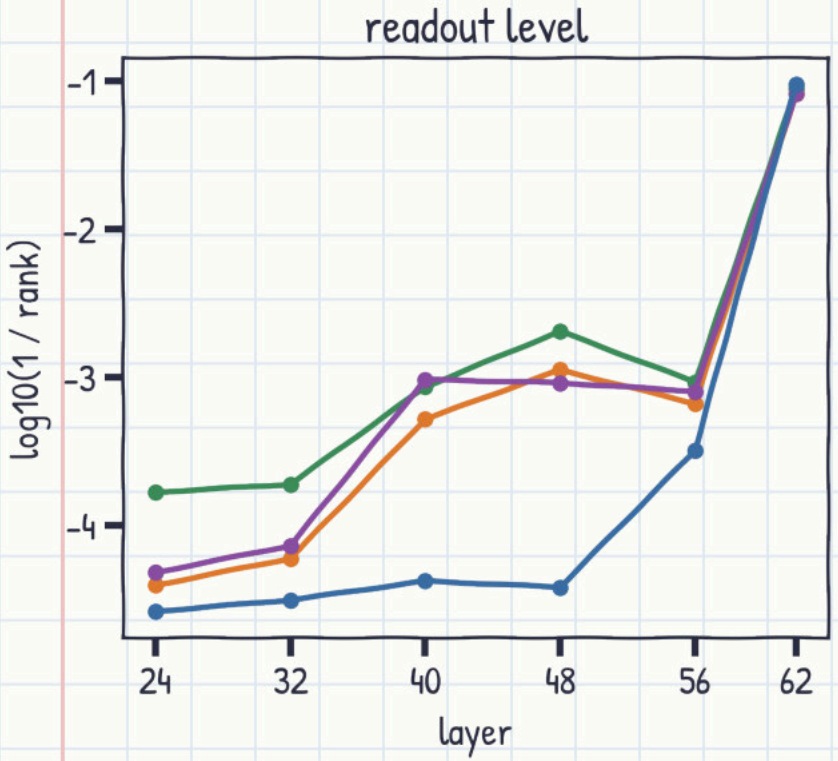
Jacobian lenses read more from **L40**

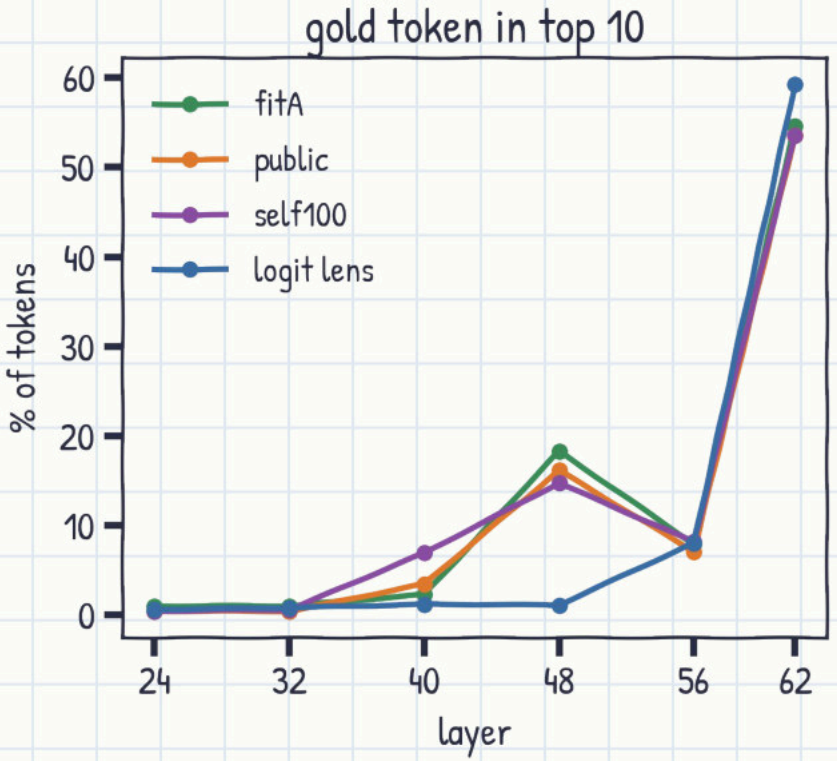
Everything jumps at **L62** (rank ~10) (Stage 10, logit L62 -0.93, §6.1)

Stage 10

Which lens reads best?

all papers · true title





Middle layers:

fitA best; at L48 15–18% top-10 vs 1% for logit

(Stage 10 X3, 9,108 papers, §6.9)

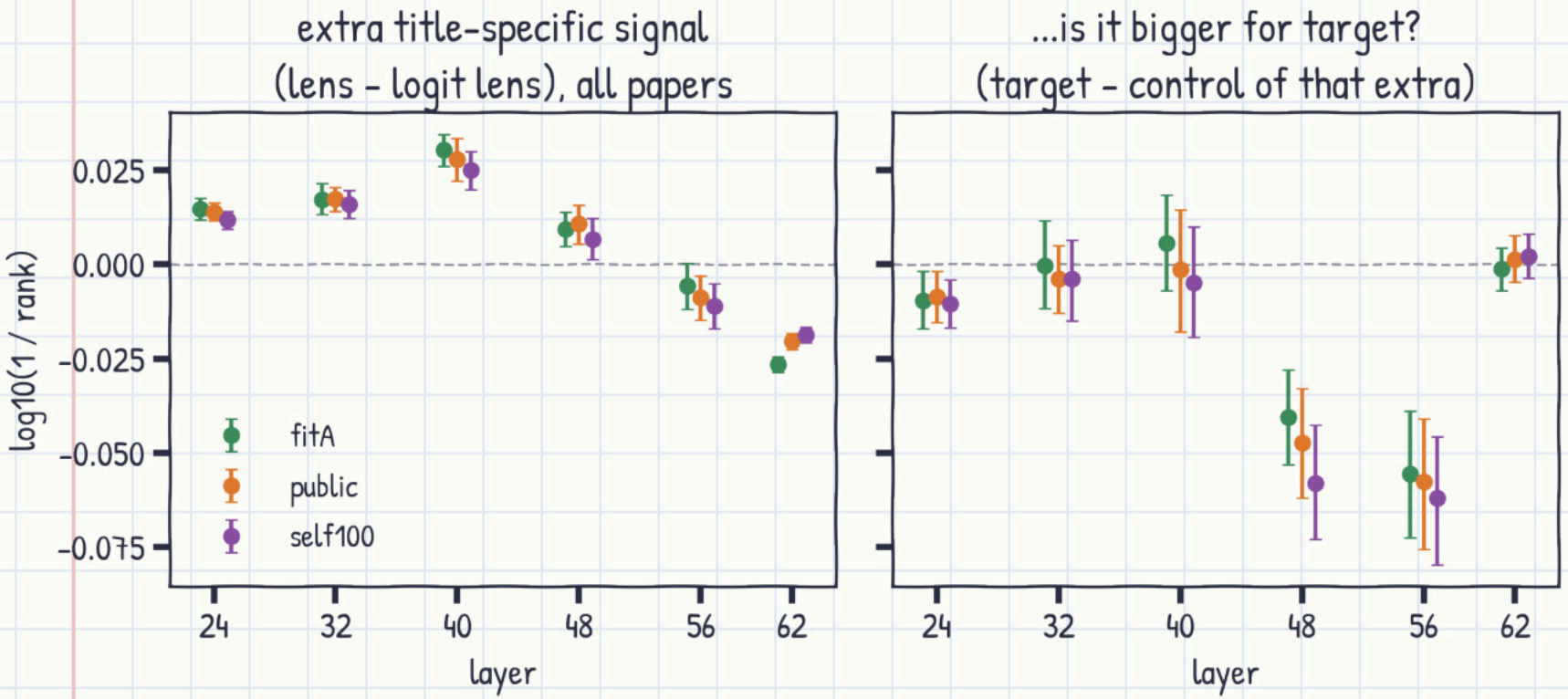
L62: logit lens best (59%)

(same, L62)

Stage 10

Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



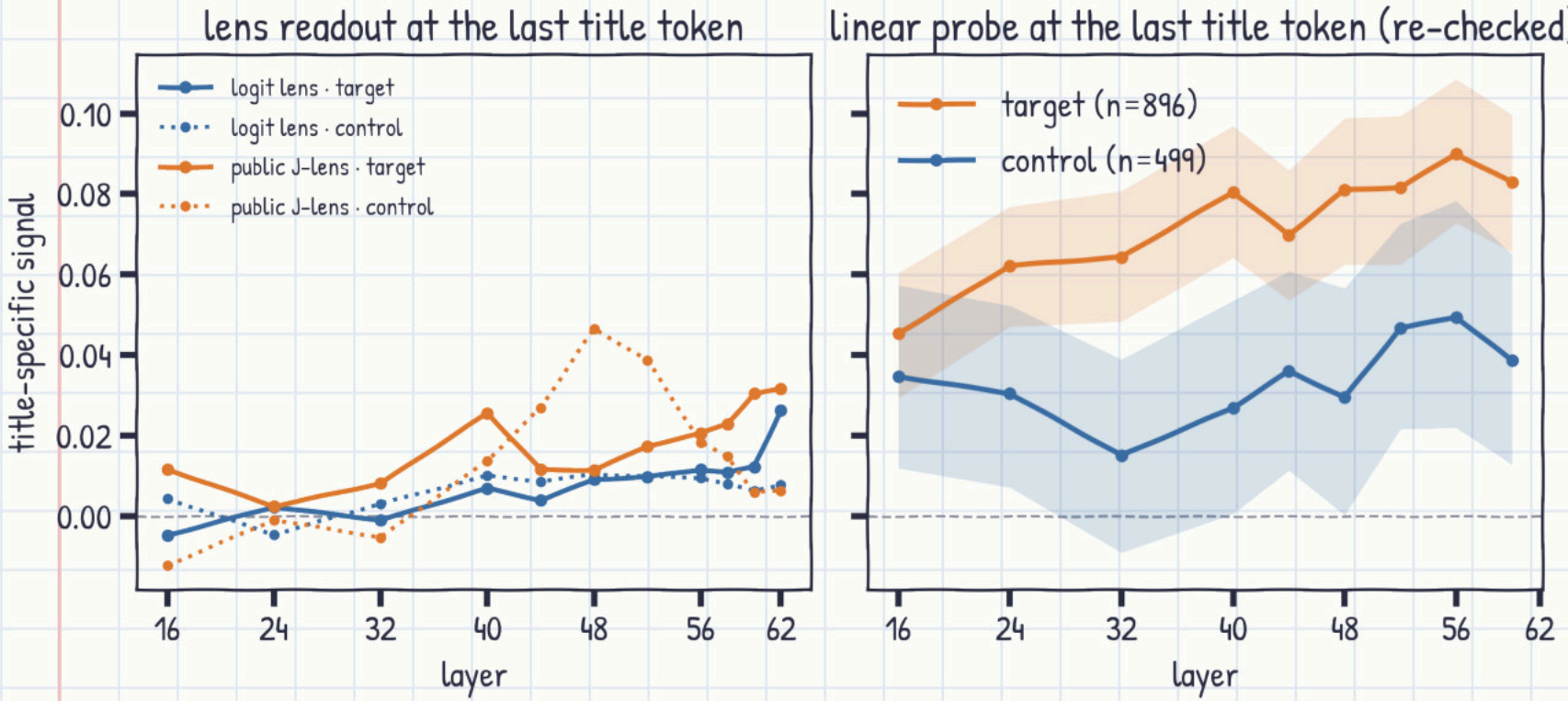
Extra signal: **+0.01 to +0.03** (Stage 10 X3, L24–L48, 9,108 papers, §6.9)

vs ~0.44 at L62 (same, L62)

Not bigger for target
(even negative at L48–56)

Read early: a probe finds it at the title

title-specific signal at the last title token · lenses: all 9,108 papers · probe: re-checked, title reuse removed: 1,863 papers (896 target, 499 control) (Stage 11 E4; Stage 13 E4b)



lenses: **~0.01–0.03** (Stage 11 E4, 4,621 + 2,220, §6.11)

probe: **0.05 -> 0.08** (Stage 13 E4b, 1,863 papers, first name token only)

target > control by **+0.03–0.08** (E4b, 896 vs 499, within bin)

re-checked: team folds, title reuse removed

scores the first name token only (mostly an initial); subtracts false_0 only

read early (title, from L24), **spoken late** (names, L57–61)

Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

Stages 1–9 · 20–200 papers

What we saw

- Self-100 fits the final distribution better; author ranking not better
- J-Lens − Vanilla ≈0 (7.4: +0.002; Stage 8: −1.4 pp)(50 / 200 papers)
- Stage 9: only Fit A fitted, stability not tested

Stages 10–11 · 9,108 papers

What scale showed

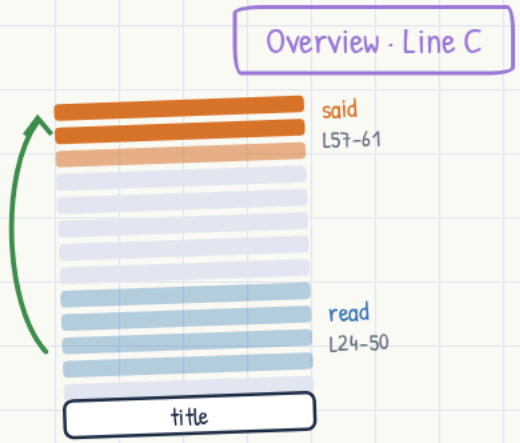
- Jacobian lenses: top-10 rate 15–18% at L48 vs 1% for logit lens(9,108 papers; X3_means.csv)
- Extra title-specific signal only +0.01–0.03(L24–L48; X3_means.csv), not target-specific
- At L62 the plain logit lens is best
- Linear probe at the title token: +0.05 → +0.08 (re-checked) (E4b, scores only the first name token; e4b_trainstd_compare.md), paper-specific

Line B: the information is there, but not in a form the lenses read

Both phases agree: no lens reveals hidden paper knowledge in middle layers. Scale adds the positive side: a trained probe can decode it at the title, so the limit is the readout, not the absence of information.

Line C

When does it appear, and when is it used?



small samples asked

Which layer peaks? (L56 kept standing out)

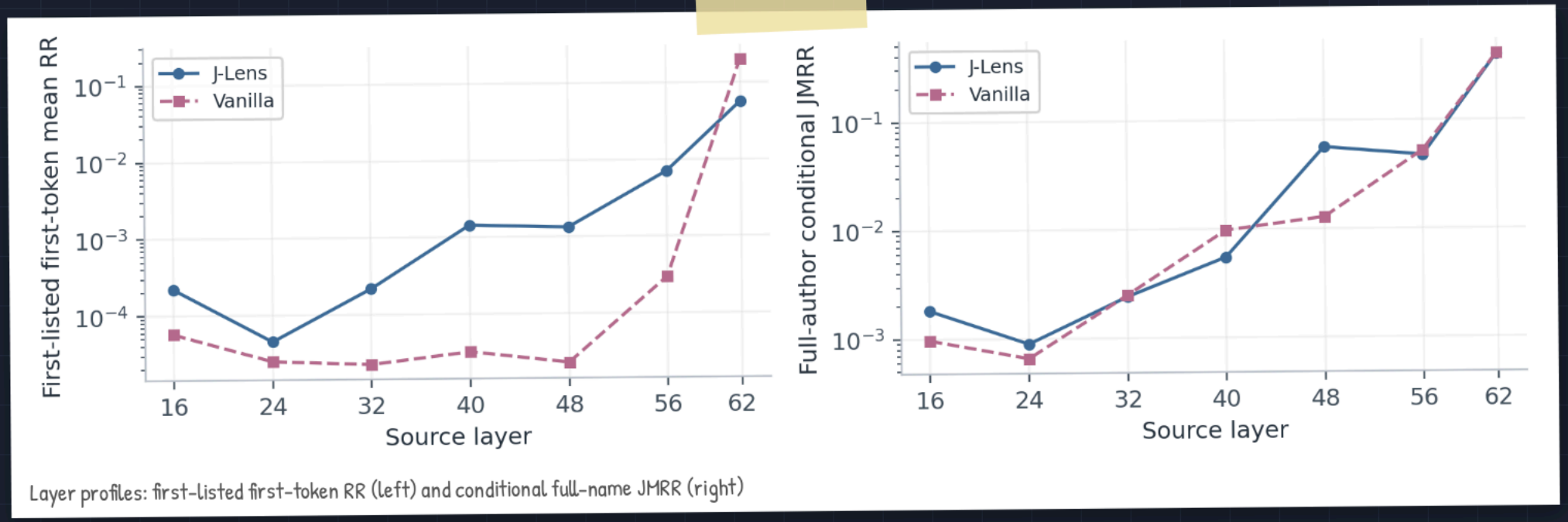
→

the full cohort answered

Readable late (L57–61) but read from the title early (L24–50)

First readout: is any author signal there?

20 papers, 93 authors (A8 / B8 / C3 / D1); ~91 s per paper, ~29.5 GiB GPU(351_stage4_readout.log)



Layer profiles: first-listed first-token RR (left) and conditional full-name JMRR (right)

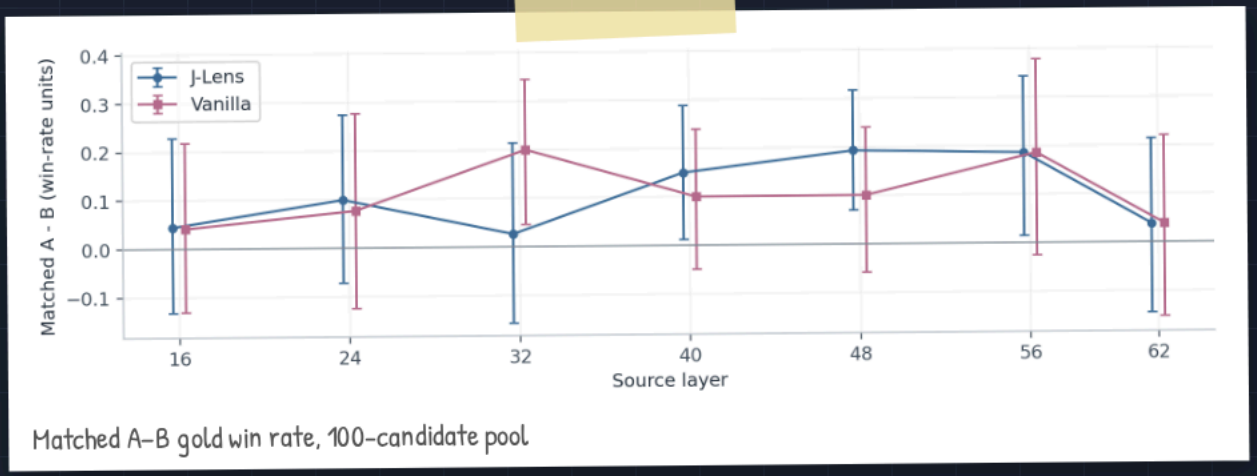
- The pipeline works and J-Lens rises above Vanilla in middle layers
- L56 first-author mean RR 0.0070, median only 0.0009(20 papers; audit_report.md table 1)

One paper = 71% of the L56 mean

Dropping it: 0.0070 → 0.0021.(same file) A layer peak can be a single easy paper, so every later stage adds controls.

Gold vs 100 real false authors

50 matched (same field, token count, year) + 50 random names per paper; 75,000 scored pairs(candidate_pairwise_scores.csv)



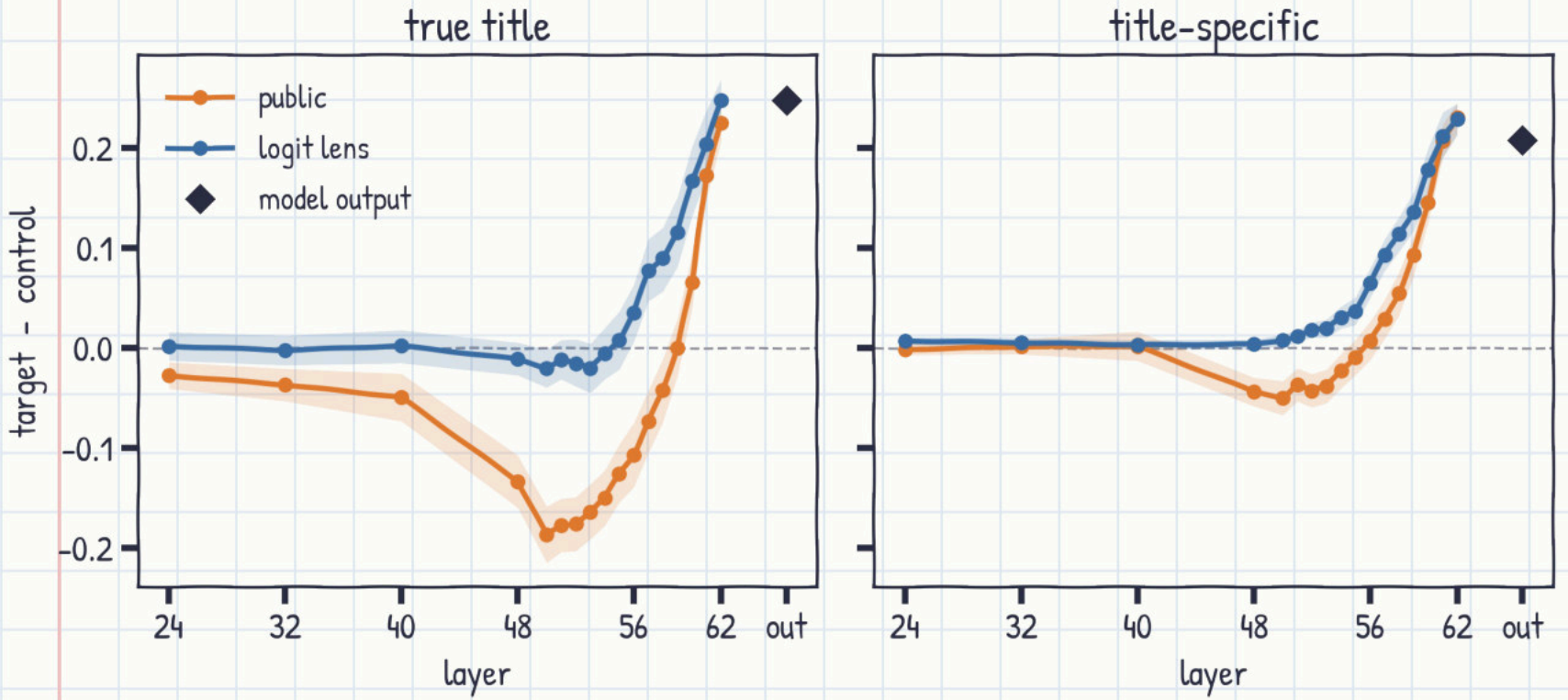
Matched A–B gold win rate, 100-candidate pool

- J-Lens A–B: L40 +0.15, L48 +0.19, L56 +0.19 (CIs above 0); L62 +0.04 (spans 0)(20 pairs, matched_ab_bootstrap.csv)
- 7-layer integrated: +0.105 [−0.012, +0.229](20 pairs; recomputed 10-30)
- 156/5,000 false names share the gold first token (ties) (quality.json)
- 100 names → only ~46 distinct first tokens(mean of 50 papers; candidates csv)

Gold beats real names in middle layers for A more than B. But the same names might win under any title: 7.3 tests that.

Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI



Middle layers: **no head start**

L62 / output:

+0.21 to +0.23 (Stage 10, L62 + final L63, 3,734 + 2,182 papers, §6.4)

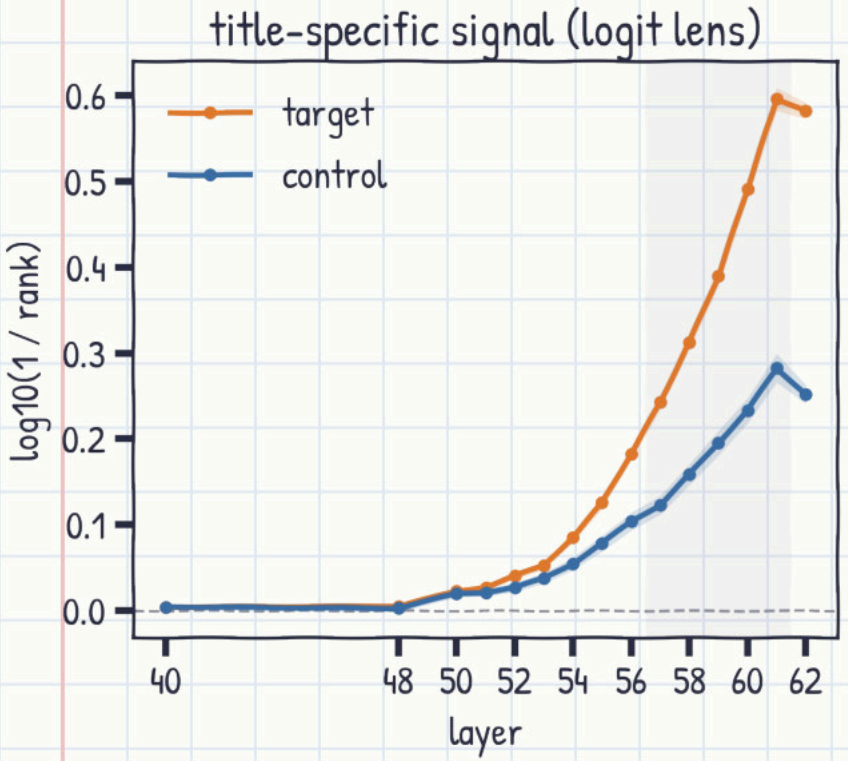
almost all title-specific

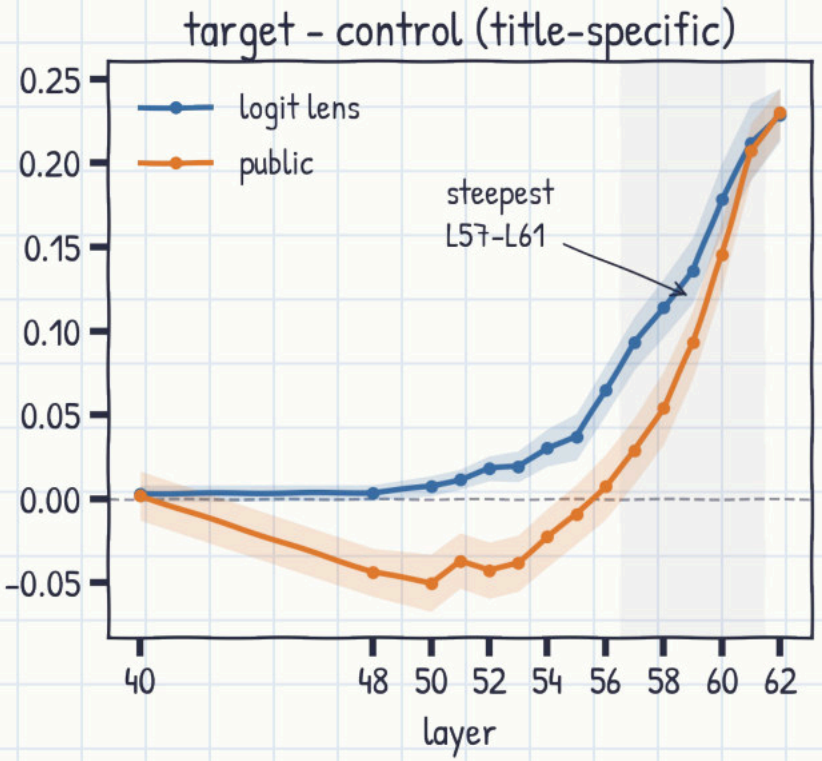
R0 only (no author written): output +0.199 vs +0.207 (Stage 10, 3,423 + 2,151 R0 papers, §6.13)

怎么算：spec = true − mean(false_0..4) 的档内 target − control · 3,734 + 2,182 篇，名字位置 · 论文 bootstrap · 出处：B_spec_target_vs_control_within_bin.csv

It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI





Logit lens: target − control

+0.007 at L50

+0.23 at L62 (Stage 10 logit lens, 3,734 + 2,182 papers, §6.10)

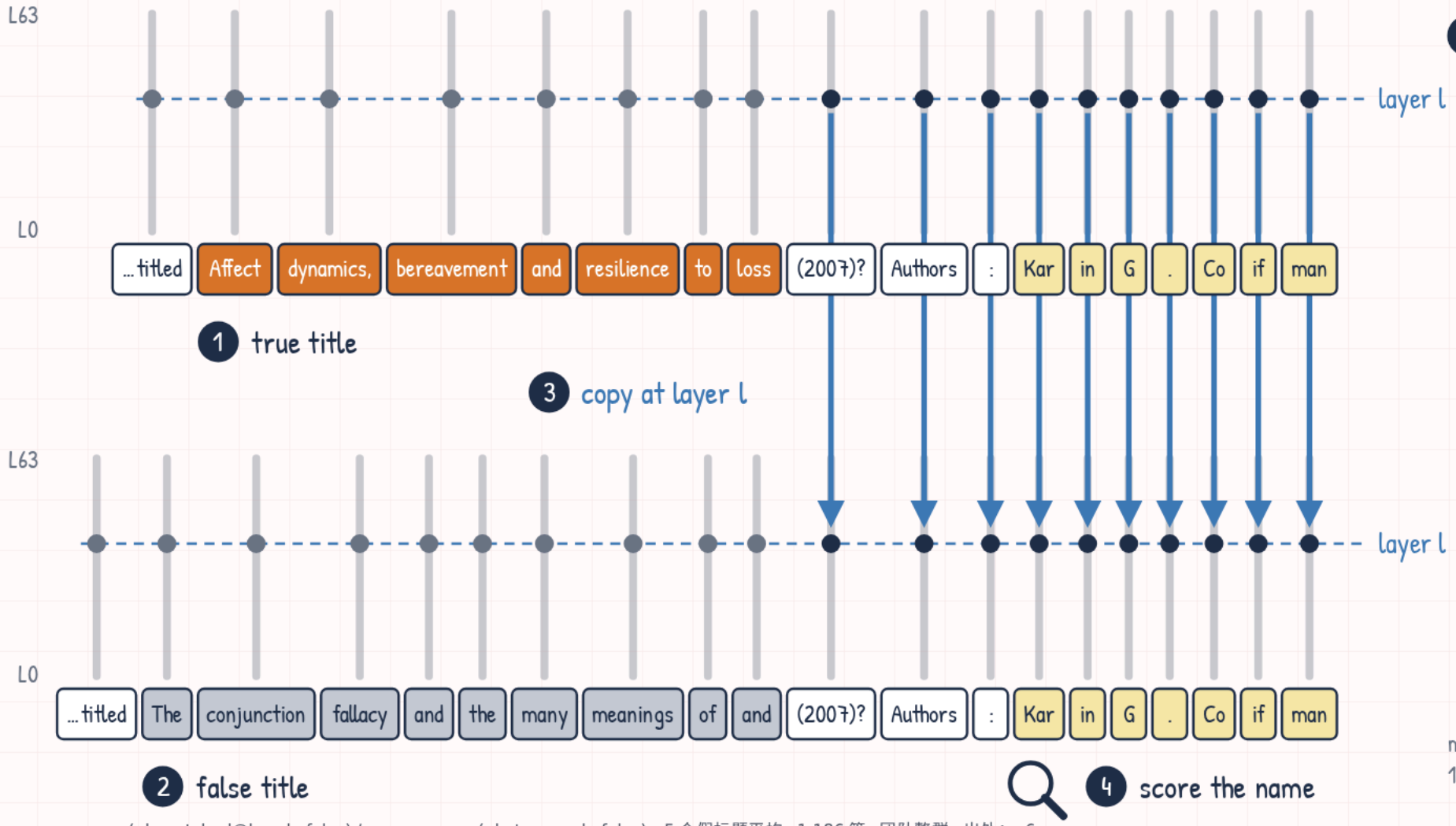
steepest rise **L57–L61**

怎么算：spec = true − mean(false_0..4) 的档内 target − control · 3,734 + 2,182 篇，L40–L62（L41–L47、L49 由 3-1 补，结论不变）· 论文 bootstrap · 出处：B_spec_target_vs_control_within_bin.csv

Replacement (patching)

Coifman, Bonanno, Rafaeli (2007)

$$R(l) = (\text{patched} - \text{false}) \div (\text{true} - \text{false})$$

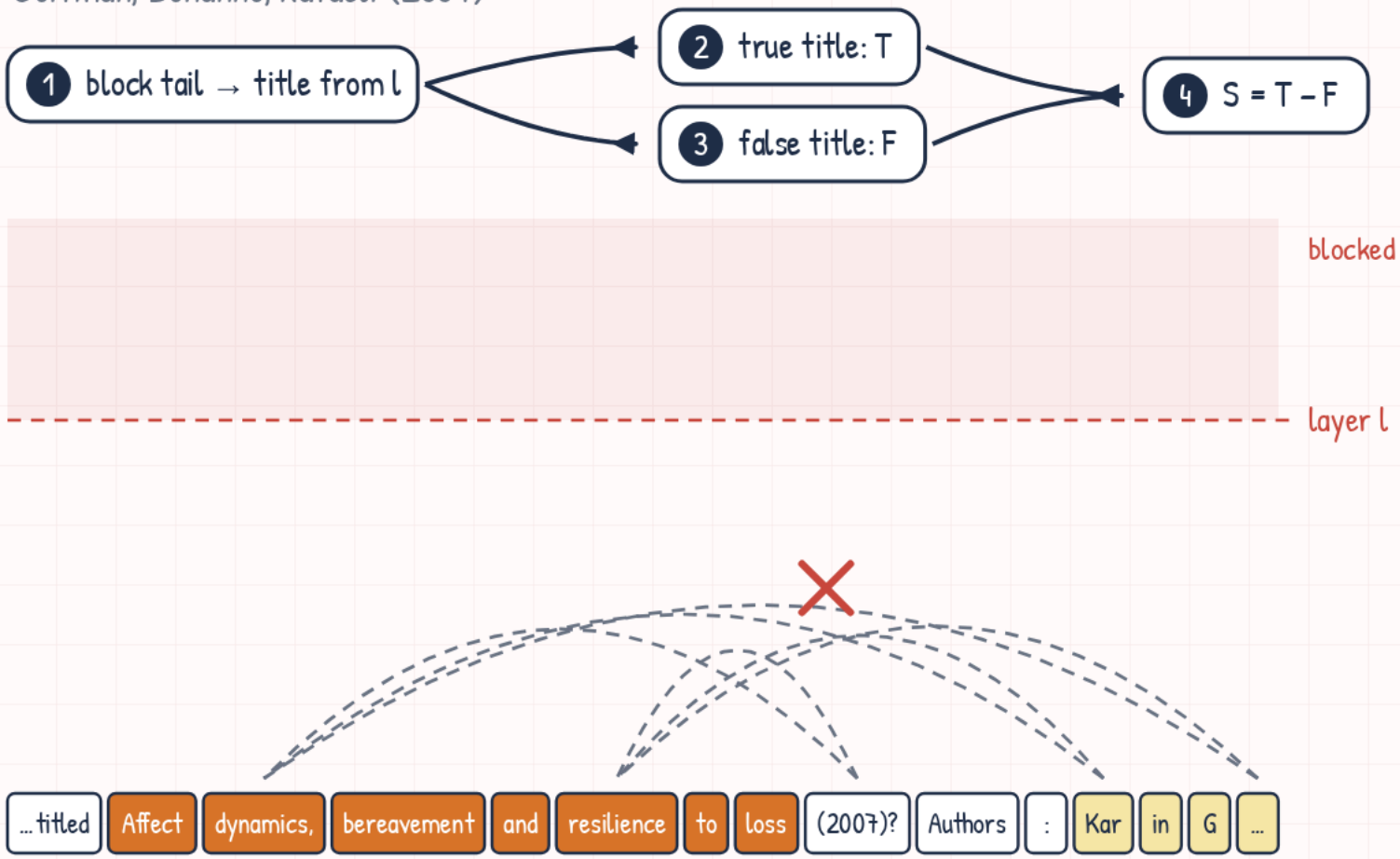


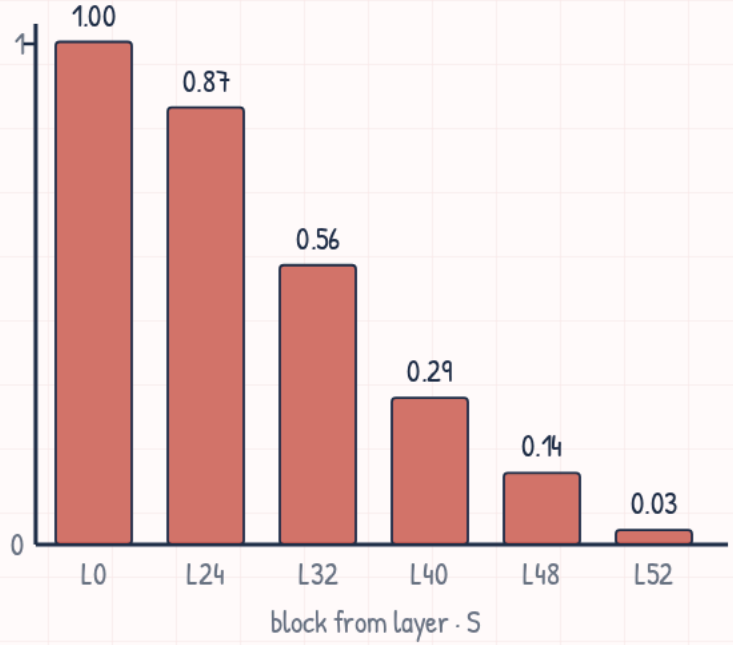
怎么算：R(l) = mean_papers(nlr_patched@l − nlr_false) / mean_papers(nlr_true − nlr_false)，5 个假标题平均 · 1,186 篇 · 团队整群 · 出处：x6_recovery.csv

R(l): Stage 13 X6, mean of 5 false titles, all 1,186 papers; 381_stage13/summary_x6/x6_recovery.csv

Knockout (block attention)

Coifman, Bonanno, Rafaeli (2007)





L52: target 0.018 · control 0.051

S(l) = T(l) − F(l)

loss as share of (true − false), 1,186 papers

S(l): Stage 13 X7, 1,186 papers (593 target + 593 control); 381_stage13/reports/x7_report.md

Stage 11: intervene, do not just read

593 + 593 papers: min(60, n_t, n_c) per citation bin, seed 11 (Stage 11 sample.parquet, §6.11)

patch uses false_0 · knockout masks all heads · all 9,108 for probe and chain (Stage 10 manifest.json)

patch

copy the true-title hidden state into the false-title run at one layer

knock out

block attention to the title tokens (all heads) from layer l to the last layer

probe

linear map from the last title token to the output state

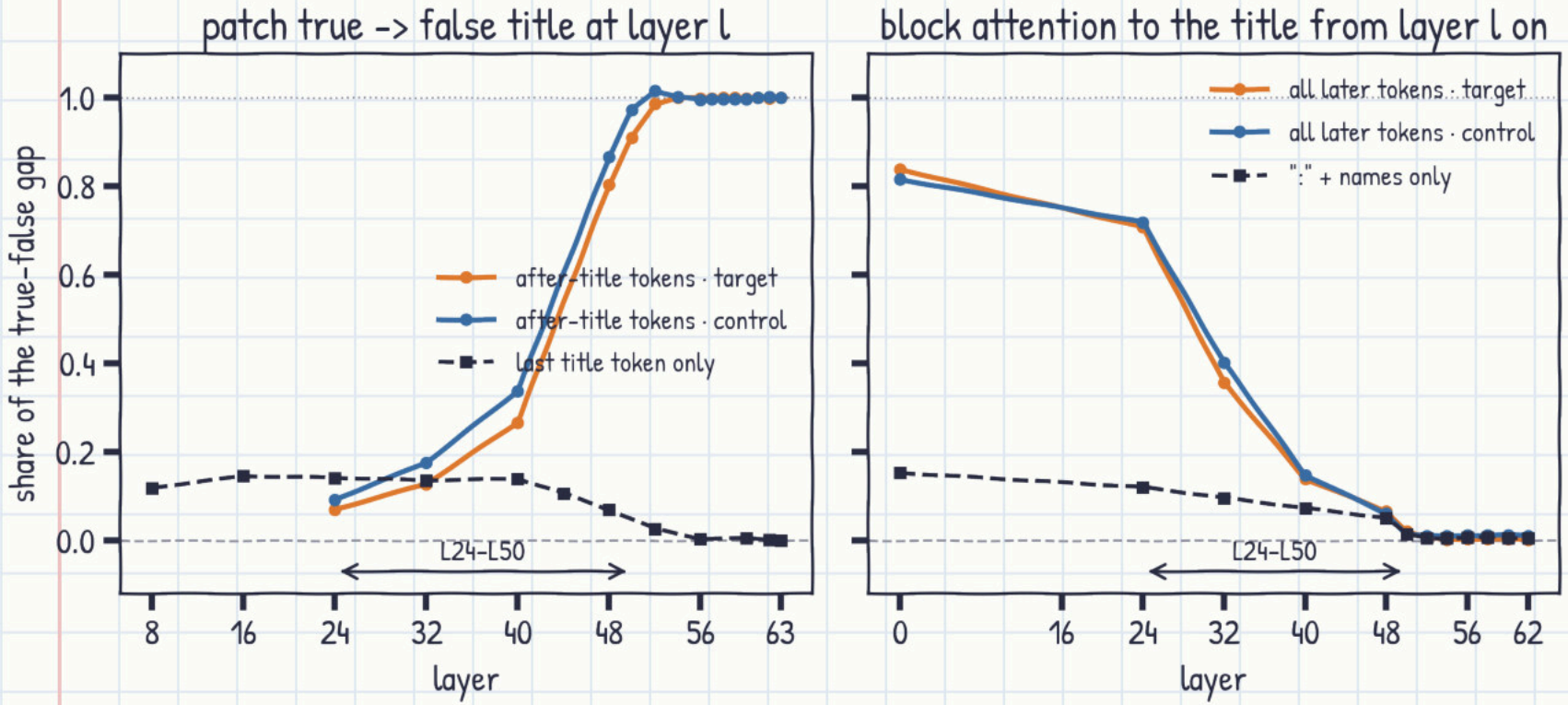
chain

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened - false) / (true - false).

The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02 (Stage 11 E1/E2, 593 + 593 papers, §6.11)



L52+: title no longer needed

the information enters the **name positions at L44–52**, partly via the tokens right after the title

same timing for target and control

怎么算： recovery(l) = mean_p[P̄_p(l) − F̄_p] / mean_p[T̄_p − F̄_p]; loss(l) = mean_p[K̄_p(64) − K̄_p(l)] / mean_p[T̄_p − F̄_p] · 593 + 593 篇，false_0 · 论文 bootstrap · 出处： E1_recovery.csv、E2_knockout.csv

Stages 1–9 · 20–200 papers

What we saw

- Stage 4 peak at L56 driven by one paper (71%)(audit_report.md)
- 7.2: A–B win-rate positive at L40–L56
- 7.3, 8: L56 title advantage; L56 became the primary layer

Stages 10–11 · 9,108 papers

What scale showed

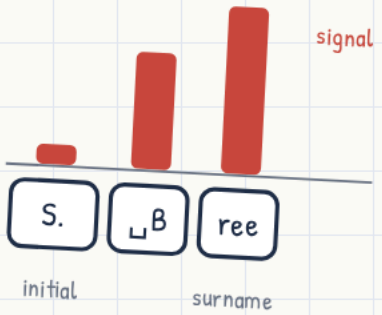
- L56 is where it *starts*; steepest rise L57–L61
- Patching: copying tail states recovers 80–87% at L48, 99.6% from L52(Stage 11 E1, 593 + 593; x6_report.md)
- Knockout: blocking the title from L52 on costs nothing

Line C: "read early, spoken late"

L56 was a real landmark, but not the peak. Interventions add what readouts cannot: the title is consumed in L24–50, and the answer only becomes readable at the name a dozen layers later.

Line D

Which token carries it?



small samples asked

First token or the rest of the name?

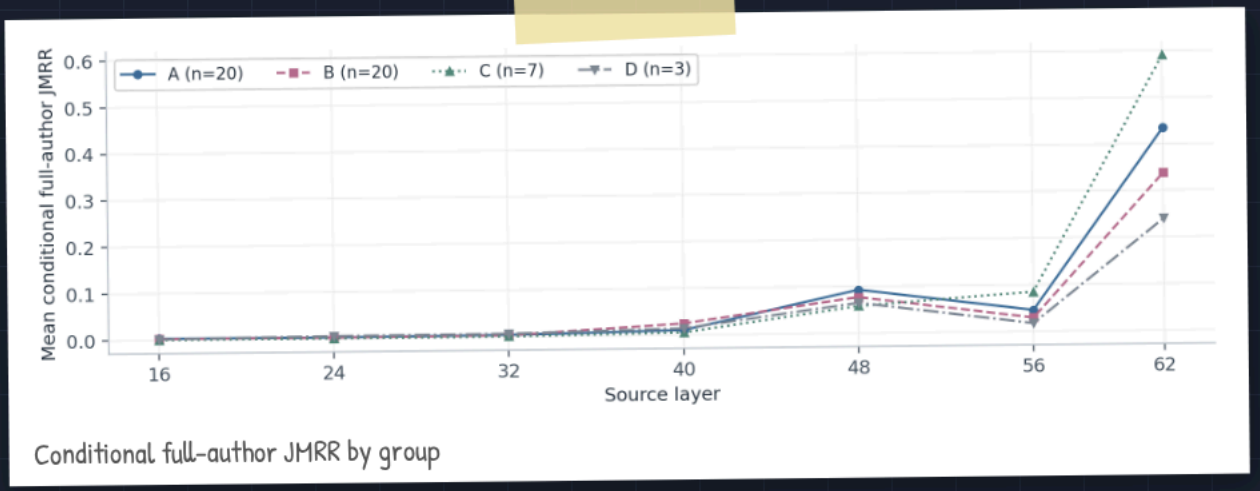
→

the full cohort answered

Depends on name format; the information enters the name positions at L44–52, partly via the tokens right after the title; co-authors partly replace the title

From first token to the full name

50 papers, 227 authors, 972 name tokens(token_layer_scores.csv); every listed author scored with teacher forcing



Conditional full-author JMRR by group

Group	Integrated JMRR(integrated_group.csv)
A (20)	0.085 [0.075, 0.096]
B (20)	0.068 [0.057, 0.080]
C (7)	0.106
D (3)	0.050

L62 A–B +0.097 [0.028, 0.163]; Vanilla also +0.091(20 pairs, matched_ab_bootstrap.csv)

Positive, but each token after the first is given the correct prefix. Next: split first token from the rest (7.1).

The A–B gap sits on the rest of the name

CPU-only re-analysis of Stage 6; 10,000 pair-bootstrap draws



Integrated A–B JMRR by token position

Endpoint	A–B (7 layers) (20 pairs; layer_and_integrated_effects.csv)
First token	+0.001 [−0.011, +0.015]
Continuation	+0.021 [+0.002, +0.039]
All tokens	+0.017 [+0.003, +0.030]

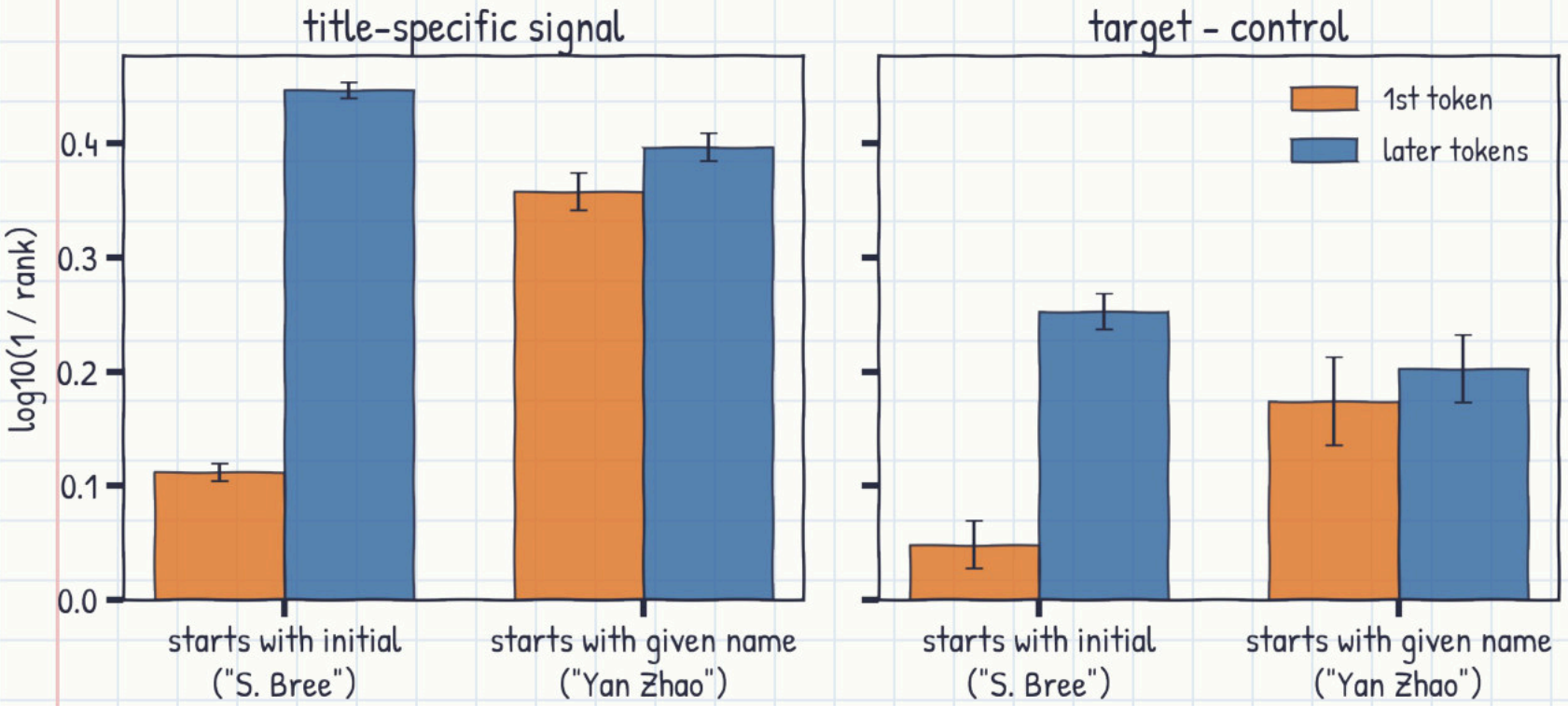
At L62: first +0.014 (12/20 pairs) vs continuation +0.121 (15/20) (layer_position_effects.csv)

With a title contrast (addendum 13 C-1): holds at the output (+0.244) and at L62 (+0.278 logit, +0.317 public J-lens); the public J-lens 6-layer mean is not significant (+0.077 [−0.021, +0.180]) (20 pairs, rest of the name; c13_C1.md)

Completing a name is easier than starting it. A continuation gap cannot show unaided recall. **Open:** is this about initials vs surnames?

First token or the rest of the name?

model output · split by name format

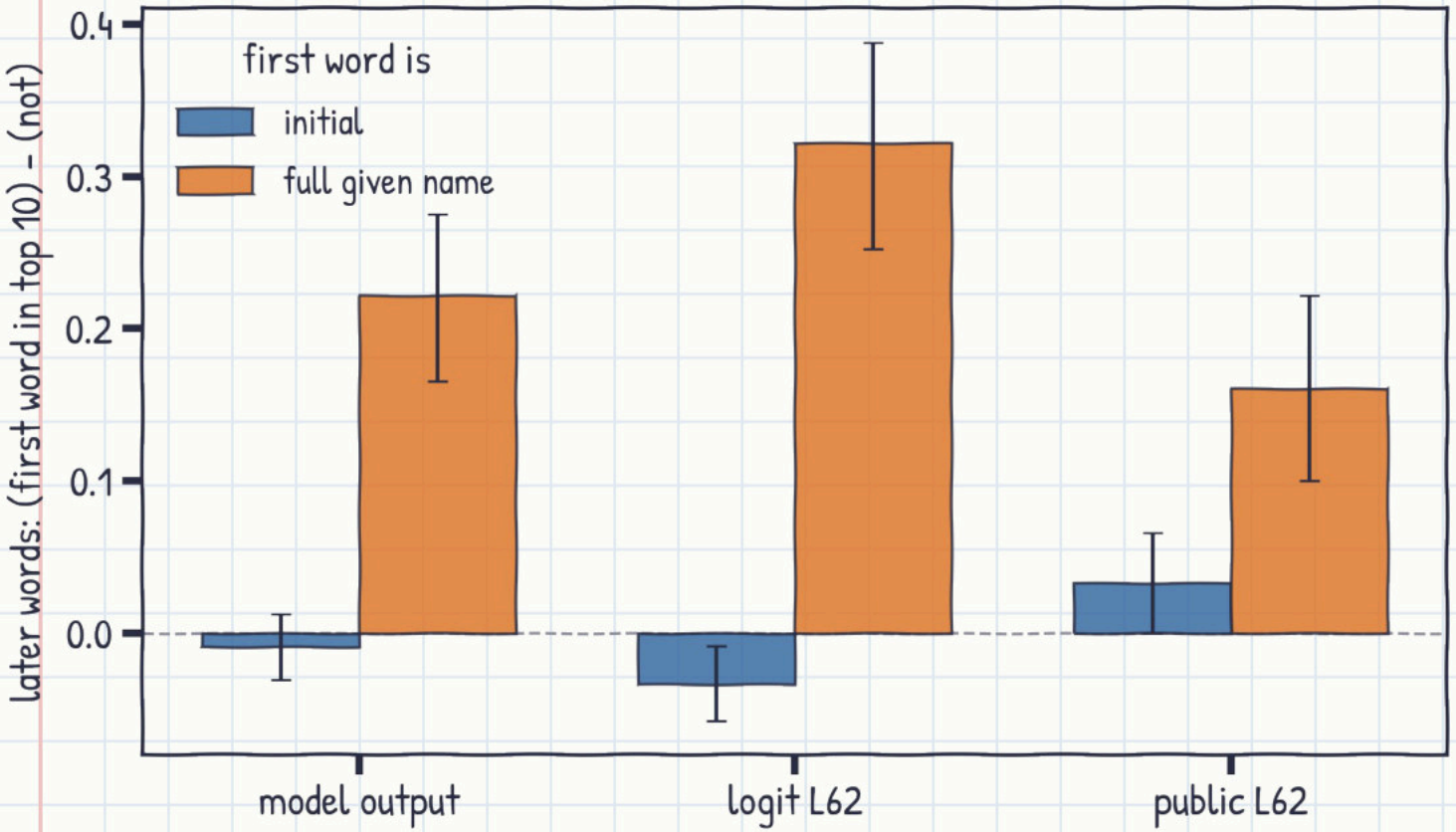


Initial: carries little; signal is on the surname

Full given name: carries about as much as the surname

Knowing the given name helps with the surname

true title · later words when the first word is / is not in the top 10



Given name in top 10

surname +0.22 (output) (Stage 10, 8,876 authors, §6.7)

+0.32 (logit L62) (same)

Initial in top 10: no help